

Smartphone-based Digital Phenotyping as a Scalable Approach to Measure Sleep Health in Emerging Adults

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INTRODUCTION

- Sleep is a critical determinant of cardiometabolic health. Smartphone-based passive sensing offers a scalable alternative for real-world sleep monitoring.
- We present findings from the 3E Smartphone Substudy, a large digital phenotyping cohort that collected over 4 billion smartphone sensor records over eight days from 587 emerging adults.

OBJECTIVES

1. **Data Completeness:** Compare the proportion of valid sleep days captured by passive smartphone data versus active self-reported sleep data
2. **Agreement:** Evaluate concordance between smartphone-derived and self-reported sleep duration

METHODS

N: 587 emerging adult college students at two public universities in California

Data Collection:

- **Smartphone sensor data** (e.g., accelerometer, GPS, activity recognition, screen state) over eight days.
- **Self-reported** sleep and wake times from the previous day over the same 8 days.

Data Analysis:

- Nightly smartphone-derived sleep estimates were captured using **screen state** and **motion-based activity recognition** levels.
- **Daily primary sleep period** was the longest contiguous interval of device inactivity (screen off and inactivity) from 8pm-12pm the following day.
- **Mixed-effects Bland-Altman analyses** were used to assess agreement between smartphone-derived and self-reported sleep duration.

RESULTS & KEY TAKEAWAYS:

1 **Smartphone passive sensing** yielded a **higher proportion of valid sleep days** than self-reported data, demonstrating more complete data with **minimal participant burden**.

2 **Self-reported sleep** was, on average, **1.74 hrs longer** than smartphone-derived sleep. However, **wide 95% confidence limits of agreement (-3.87 to 7.34 hrs)** indicate **substantial day-to-day disagreement** between methods.

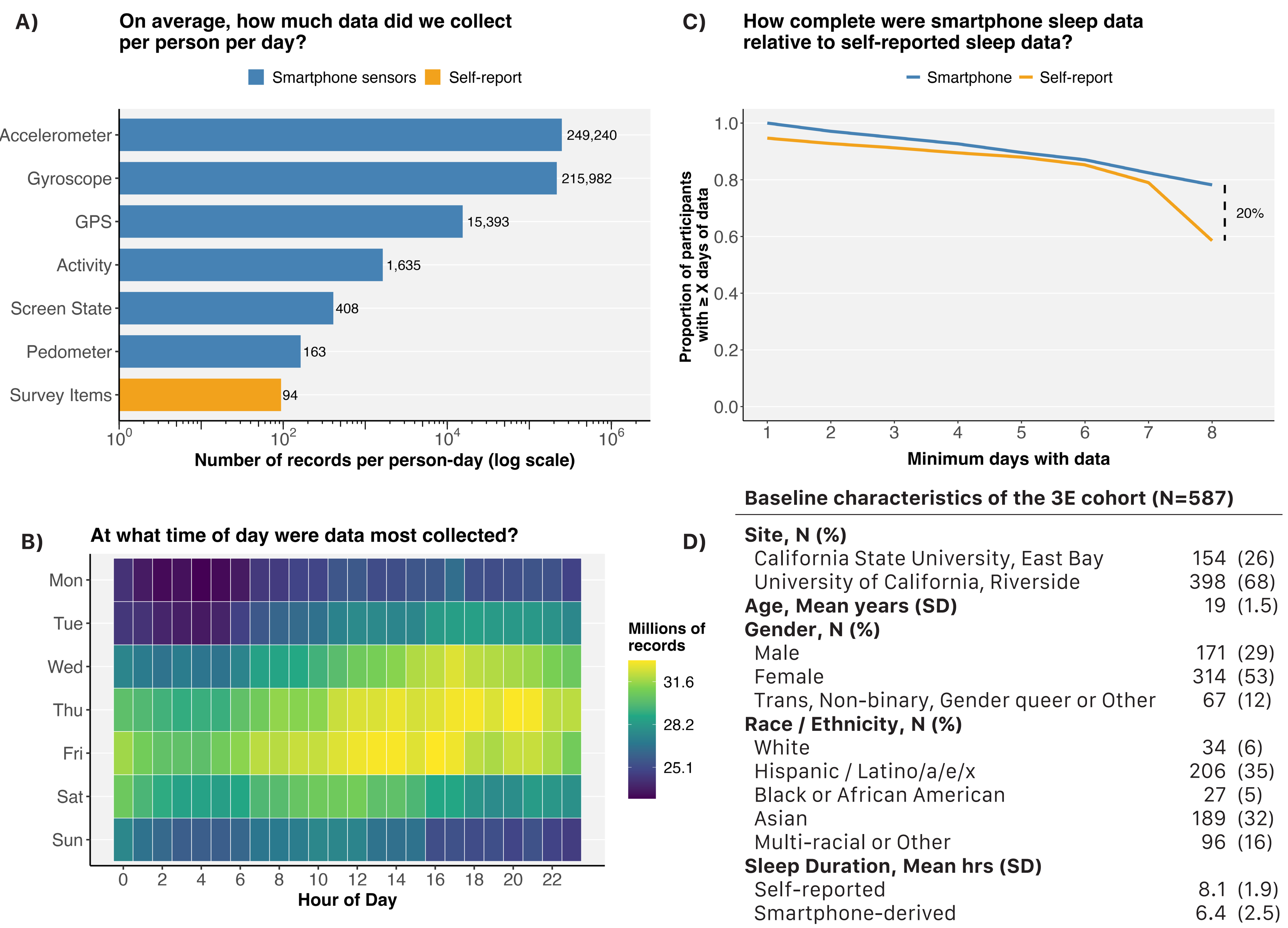


Figure 1. Overview of smartphone sensor data collection from 587 participants. A) Average number of records collected for each data stream per participant per day. B) Heatmap showing temporal patterns of data collection by hour and day of week. C) Data completeness curves comparing smartphone sensor data to self-reported survey data. D) Baseline characteristics of the cohort; 35 participants (6%) had missing characteristics for site, age, gender, and race and were counted as missing in the totals and percentages above.

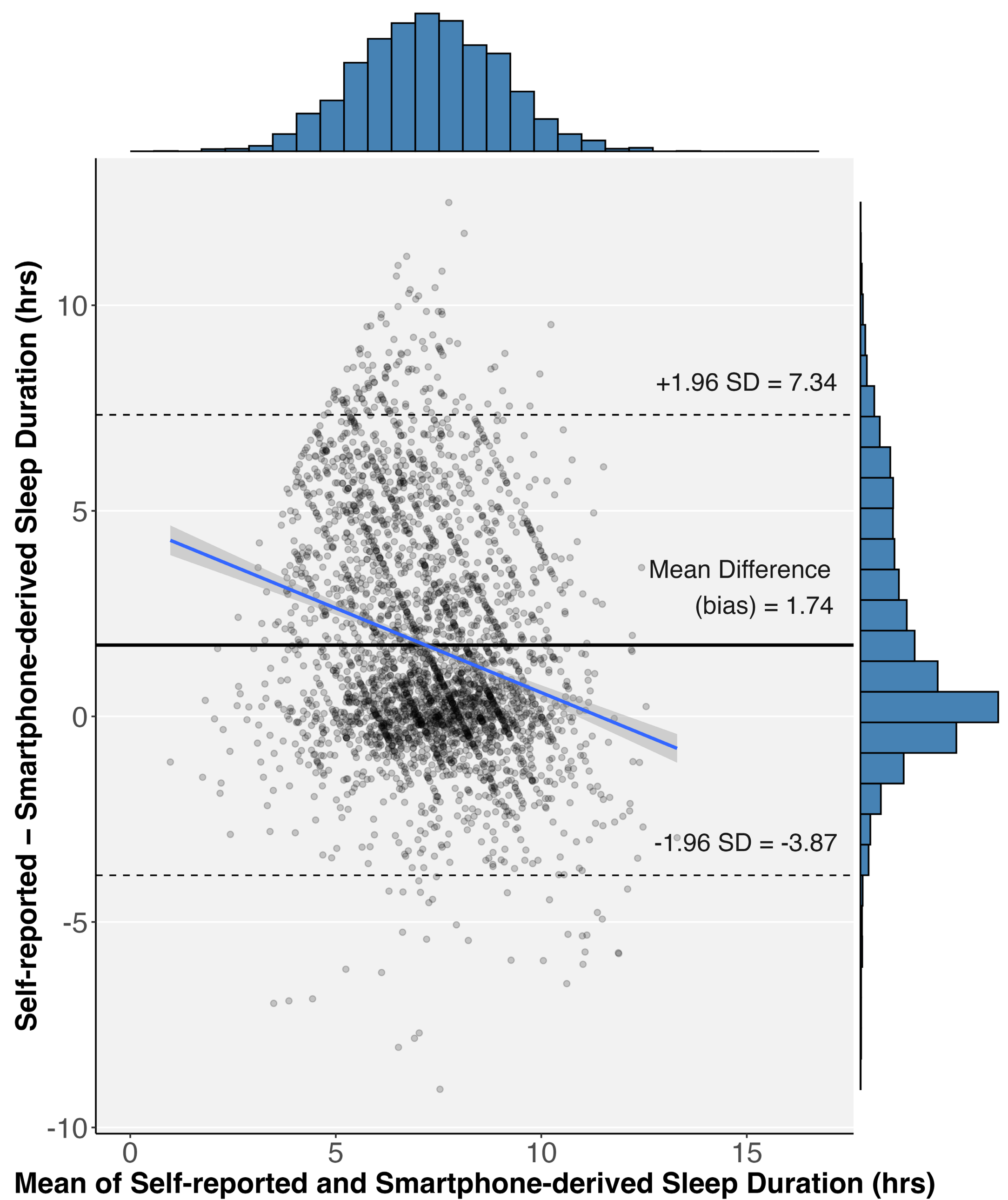


Figure 2. Mixed-effects Bland-Altman plot comparing self-reported and smartphone-derived sleep duration. Each point represents a daily observation per participant. The solid line shows the estimated mean difference (bias) from a random-intercept linear mixed model. Dashed lines indicate the 95% limits of agreement based on total (between- and within-person) variance. Histograms in the margins display the distributions of mean sleep duration (top) and method differences (right).

CONCLUSION & NEXT STEPS

- Smartphone sensing offers a feasible, scalable, and extremely low-burden way to capture high-resolution behavioral data in naturalistic settings and diverse populations.
- While smartphone-derived sleep estimates showed a reasonable average bias relative to self-report, wide limits of agreement indicate substantial day-to-day disagreement at the individual level. Smartphone sensing

may be useful for estimating population-level sleep patterns, but caution is warranted when interpreting individual-level estimates.

- Smartphone sensors capture behavioral proxies of sleep rather than physiological sleep itself.
- Future work will refine smartphone-based algorithms to better understand and reduce sources of measurement disagreement.

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